Topic 6 Review [84 marks]

1. [Maximum mark: 4] 24M.1.HL.TZ1.9

Outline **two** operating system resource management techniques. [4]

Markscheme

Award ***[2 max]***

Award ***[1]*** *for technique and award* ***[1]*** *for description.*

Scheduling;
Selecting a process (from a ready queue) and allocating CPU to this process for execution;

Time Slicing;
Allocating slices/ fractions of time to each process to enable programs to run simultaneously;

Multitasking;
Allowing a user to perform more than one computer task at a time;
to enable more than one application to run at the same time/ simultaneously;

Virtual memory;
A section of secondary storage set up to act as a primary storage;

Paging;
A memory management mechanism used to swap processes from secondary storage to primary storage as they are needed, (in the form of equal sized pages);

Interrupt;
A signal emitted by hardware or software to require the processor to stop its

current process (if it is of lower priority) and action the task related to the interrupt;

Polling;
The constant checking the status of devices to see if they need the attention of the CPU;

Note: *Accept other appropriate description.*
No mark for description if the name of the technique is not correctly identified.

Examiners report

Most of the students were able to achieve at least two marks out of four here. Very few students were able to describe the technique using specific key terminology.

2. [Maximum mark: 15] 23N.1.HL.TZ0.12

Input devices that detect cars approaching a crossroads are connected to a microprocessor.

(a.i) Identify **two** types of sensor that can be used to detect approaching cars. [2]

Markscheme

Award ***[2 max]****.*
Electromagnetic sensors;
Pressure sensors;
Infrared sensors;
Ultrasonic sensors;
Microwave sensors;

Inductive loop detectors;
Acoustic sensors;

Examiners report

Terminology is not always used in an appropriate and competent way and these questions are often approached by providing superficial common knowledge

The vast majority of students correctly identified two types of sensors that can be used to detect approaching cars.

(a.ii) Outline why sensors are appropriate input devices in this situation. [2]

Markscheme

*Award **[2 max]**.*
Sensors are appropriate because they automatically sense the presence in the given environment;
without the need for human operation/sensors can detect 24x7 without losing efficiency;

Sensors are appropriate because only presence of a car required;
this can be reliably detected by sensors/data is sent to CPU quickly enabling quicker reaction times;

Sensors are appropriate because it saves staffing the crossroads;
so, costs are reduced;

(b) Suggest the type of memory that could be used to store the control program in the microprocessor. [2]

Markscheme

Award ***[2 max]***.
ROM;
Because it is non-volatile;

PROM;
can be programmed only once and is not erasable;

EPROM;
Can be reprogrammed in case of change in traffic flow;

EEPROM;
can be erased one byte at a time (rather than erasing the entire chip)/allows flexible process of reprogramming;

Examiners report

A mixed set of responses was seen for this question.

The traffic lights at the crossroads are also connected to a microprocessor. A person who wishes to cross the road presses a button at a traffic light. This causes an interrupt.

(c.i) Outline what is meant by an interrupt. [2]

Markscheme

Award ***[2 max]***.
An interrupt is a signal sent to the processor;
that halts the current process;

Examiners report

Many students provided answers that were vague or too general.

(c.ii) Explain how the microprocessor can deal with this interrupt. [3]

Markscheme

*Award **[3 max]**.*

The microprocessor responds to that interrupt with an interrupt handler/an ISR (Interrupt Service Routine)/a routine which instructs the microprocessor on how to handle the interrupt;
the routine can send a signal to output transducers;
to immediately turn the traffic lights to red to stop the cars;

The microprocessor receives an interrupt request and passes control to the interrupt handler;
it may wait a certain amount of time before sending the signal to output transducer;
to turn the traffic lights to red to stop the cars;

Cameras are installed on the top of the traffic lights at the crossroads.

(d.i) Outline **one** benefit of monitoring the traffic with cameras. [2]

Markscheme

*Award **[2 max]**.*

Cameras improve traffic flow;
by controlling the movement of vehicles and determining traffic light timing;

Traffic cameras help to reduce accidents;
by controlling the movement of vehicles/identifying drivers who speed;

Cameras improve safety/have a preventive/disciplining effect;
can be used to identify defaulters and to take legal action/make them pay fine;

Installation and deployment of automated traffic cameras often pays for itself in the long run;
is still far less expensive than physical policing/traffic cameras can operate tirelessly 24/7;

Traffic cameras can be deployed with very little cost to infrastructure;
as they can run independently from the electrical grid only on solar power;

Examiners report

Most students provided generally correct answers. Students lost marks by not providing answers with sufficient clarity.

(d.ii) Outline **one** concern about monitoring the traffic with cameras. [2]

Markscheme

*Award **[2 max]**.*
Privacy is an issue;
as traffic cameras enable monitoring the activities of people (drivers, pedestrians);

It can be costly;
to install/maintain a large number of cameras and monitoring systems;

Cameras can be vulnerable to damage;
For example, wetness, moisture;

Cameras can be vulnerable to misuse;
for example, criminals/hackers might have worked out ways to disable or disconnect cameras;

3. [Maximum mark: 4] 23M.1.HL.TZ1.4

Sensors that take readings of the levels of different pollutants have been installed at a number of sites along a river. Each reading is sent to a central computer, where it is processed and analysed.

(a) Define the term *interrupt*. [1]

Markscheme

*Award **[1 max]***
(Software) interrupt is a signal emitted by software (when an application program terminates/ requests certain services from the OS/ a process (or an event) needs immediate attention;

(Hardware) interrupt is a signal created by some action taken by a hardware device and sent to the CPU to stop its current activity;

Examiners report

Many good and comprehensive answers were seen.

(b) Describe how polling could be used in this situation. [3]

Markscheme

*Award **[3 max]***
*Award **[1]** for CPU checking status of each sensor to see if it needs attention*
*Award **[1]** for identifying/ sampling their status by a program (running on the computer)*
*Award **[1]** for operating at the same periods of time*
*Award **[1]** for relation to given situation*

The computer could continually interrogate all connected sensors along the river;
to identify if their status has changed/ they require to transmit data;
the readings from each sensor along the river could be taken at regular intervals;

in an order determined by the program running on the computer/ at a frequency determined by the program running on the computer;
ensuring that the frequency of polling increases in the parts of the river where the measurements are deemed to be higher/ in specific times when the quality of the water is known to be worse/ in times of lower use of the (central computer) processing units;

Examiners report

Many candidates scored full or nearly full marks for this question. Some candidates confused the term 'polling' in computing with focus groups in which people are polled about the river pollution.

4. [Maximum mark: 3] 22N.1.HL.TZ0.7

Explain how increasing the size of the central processing unit (CPU) cache improves the performance of a computer. [3]

Markscheme

*Award **[3]** max.*
Commonly used instructions from a program are stored in cache memory to be accessed by processor (faster access than from RAM);
Increasing the size of cache memory means that more instructions can be quickly found/accessed by the processor;
therefore, program execution is made faster;

Examiners report

This question was well answered with the vast majority of candidates gaining some marks, some of whom achieved full or nearly full marks.

5. [Maximum mark: 3] 21N.1.HL.TZ0.8

Explain how an operating system manages peripherals. [3]

Markscheme

Award ***[3 max]***

OS keeps tracks of all peripheral devices (the I/O controller) / decides which process gets the device when and for how much time / allocates and de-allocates devices;
OS works with device drivers and the basic input/output system (BIOS) to perform hardware tasks
/ the necessary drivers (for every peripheral) are built into the OS and/or when a new peripheral is added software / device driver provided by a hardware maker is installed into the operating system (to tell the computer's OS how to work with the peripheral/hardware) (because without a device driver the OS would not be able to communicate with this peripheral device);
A device driver translates the OS's instructions into a language (analogue signals) that the device can understand;
There are various types of device drivers for peripherals (such as keyboards, mice, disk drives, controllers, printers, graphics cards, ports, etc.);
Device drivers run in the OS kernel space (in the part of the OS that directly interacts with the physical structure of the system) (and implement functions such as open, close, read, write);
The running application/user's program can make a call to device driver functions which provide an interface between user space and kernel space;

Example answer:
An OS manages peripherals via their respective drivers;
For example, sound card drivers;
are necessary so the OS knows exactly how to translate the 1s and 0s (that comprise the MP3 file) into audio signals;
that the sound card can output to headphones / speakers;

Examiners report

A well answered question. Many comprehensive answers were seen.

6. [Maximum mark: 3] 22M.1.HL.TZ0.8

State **three** operating system resource management techniques. [3]

Markscheme

Award ***[3 max]***

scheduling;

policies;

multitasking;

virtual memory;

paging;

interrupt;

polling;

Examiners report

A well answered question.

7. [Maximum mark: 1] 21M.1.HL.TZ0.9

Identify **one** advantage of using a dedicated operating system on a mobile phone. [1]

Markscheme

Award ***[1 max]***

Makes maximum use of the available (limited) memory to provide the

features needed for the phone;
Does not waste memory space with unwanted/inappropriate features;
Ensures a higher level of security;
Makes efficient use of hardware equipment;
Suited to available hardware equipment;

Note*: Accept examples, GUI will fit right with the screen resolution /can make running the device easier to use / better suited to their audience/ it is faster than universal OS's, etc*.

Examiners report

Many candidates correctly answered this question.

8. [Maximum mark: 18] 21M.1.HL.TZ0.15

A business has a range of different computers within the organization, including laptops, desktops and file servers. Wherever possible the organization uses a common operating system on its computers.

(a) Outline **two** resource management techniques that are likely to be carried out by the operating system of a desktop computer. [4]

Markscheme

Award ***[4 max]***

Virtual memory;
to enable the hard drive to act as primary memory if required;

Paging;
scheme by which a computer stores and retrieves data from secondary storage for use in main memory;

Multitasking;
to enable more than one application to run at the same time;

Scheduling (method by which work is assigned to resources that complete the work);
to determine which process will own CPU for execution while another process is on hold.;

Policies (ways to choose which activities to perform);
according to which decisions are made about which operations should be authorized/ which resources should be allocated;

Interrupt handling;
so that urgent tasks required by parts of the system may be executed;

Polling;
to sampling the status of an external device;

Mark as 2 and 2.

Examiners report

This question was answered well by many students. Part (a) was reasonably well answered with many gaining at least two of the marking points.

(b) Outline **one** way the operating system hides the complexity of the hardware from the computer user. [2]

Markscheme

*Award **[2 max]***
Award 1 mark for the virtual machine (presented by the OS to the user, hiding all the complexities of the hardware behind layers of OS software) and 1 mark for an expansion.

The virtualization of real devices;
such as the use of drive letters for partitions to represent a virtual hard drive;

Use of icons;
to represent peripherals/to access different drives;

Examiners report

Part (b) was well answered.

Memory requirements and processor speed will vary depending on the tasks required of the computer.

(c.i) Contrast the memory requirements of a laptop computer and a file server. [2]

Markscheme

Award ***[2 max]***
The primary memory/secondary memory in a laptop will be less / smaller than would be found on a file server;
because it is a smaller device that is used by an individual, so doesn't need to store / run so many programmes simultaneously;

Examiners report

Students generally have problems with compare/contrast questions, describing the requirements individually without contrasting, as was the case in Part(c).

(c.ii) Contrast the processor speed requirements of a laptop computer and a file server. [2]

Markscheme

Award ***[2 max]***
The processor of a file server would be faster/have more cores than that of a laptop;
as it has many more resources to manage/as it needs to keep track of many workstations/printers/peripherals;

Examiners report

Students generally have problems with compare/contrast questions, describing the requirements individually without contrasting, as was the case in Part(c).

The business has decided to implement a computer-based system to switch the room lights on and off automatically. The lights will only be switched on if the level of light is below a specific reading and there are people in the room. The lights will be switched off when the room has been unoccupied for at least five minutes.

(d) State **two** types of sensor that are required to control the lighting to ensure it switches on when it is required. [2]

Markscheme

Award ***[2 max]***
Light sensor;
infra-red/motion sensor;

Examiners report

A large number of candidates scored full or nearly full marks for the questions (d)

(e) Explain how the system makes use of the data it receives from the sensors to determine when to switch the lights on. [4]

Markscheme

Award ***[4 max]***
The sensors send data to the processor continuously;
The data received from the sensors is converted to digital (using an analogue to digital convertor);
The readings are compared to pre-set values in the controller/system;
If the light level is found to be below the pre-set value **and** a presence is detected (in the room);
A signal is sent to the actuator to turn on the lights;

Examiners report

A large number of candidates scored full or nearly full marks for the questions (e)

(f) Outline how the system will prevent the lights from being switched off too quickly when it thinks the room is unoccupied. [2]

Markscheme

Award ***[2 max]***
A timer (set to 5 minutes);
If the light level is found to be below the pre-set value a timer is activated but the lights remain on;
When the timer reaches 5 minutes, the lights switch off (if no further presence is detected);

Examiners report

A large number of candidates scored full or nearly full marks for the questions (f)

9. [Maximum mark: 3] 20N.1.HL.TZ0.3

Describe how an operating system uses paging when running a program. [3]

Markscheme

Award ***[3 max]***

memory management method that uses secondary memory to increase the amount of primary memory;
transfers data blocks of the same size (“pages”);
from secondary storage to main storage when they are required;
and returns them to secondary storage when they are not;

Examiners report

Not all candidates answered this question correctly, but most of those who did, scored good marks.

10. [Maximum mark: 15] 20N.1.HL.TZ0.11

A company produces and sells domestic floor-cleaning robots.

The floor-cleaning robots can clean different surfaces like wood and carpet. The floor-cleaning robots can also avoid obstacles or stairs.

Sensors are used by the processor that controls the floor-cleaning robot so that it can move safely.

(a) Describe **two** types of sensors used in the floor-cleaning robots. [4]

Markscheme

Award ***[4 max]***

Proximity sensors/ range sensors;
Which are used to determine how close an object is to the sensor;

Optical sensors /Photocells and other photometric devices (often used in conjunction to proximity sensors);
which are used to detect the presence or absence of objects;

Tactile sensors / Contact sensors / Bumpers;
which are used to determine whether contact is made between sensor and another object;

Touch sensors;
which indicate when contact is made;

Force sensors;
which indicate the magnitude of the force with the object;

Machine vision;
which is used in robotics for inspection / parts identification / guidance (accept other uses);

Mark as 2 and 2

Award marks for other miscellaneous category of sensors which also are used in robots. For example, devices for detecting / measuring temperature / fluid pressure / fluid flow / electrical voltage / other physical properties/.

Examiners report

Mostly a well answered question.

(b) Explain the function of an output transducer in this situation. [3]

Markscheme

Award ***[3 max]***
Output transducer is a device that accepts a (digital) signal from processor;
and turns it into a physical movement;
to make the floor cleaning robot move in different direction;

Examiners report

Candidates mostly gave good responses for this question. Some candidates confused output transducers with sensors.

A computerized security system for the company's headquarters protects against unauthorized access using a swipe-card system. Each door has a swipe-card reader that is connected to the central computer. A database stores the IDs of all employees and the rooms they are allowed to access.

(c.i) Identify **one** alternative computerized method that could be used in place of the swipe-card readers. [1]

Markscheme

Award ***[1 max]***
Any biometric device (finger print, eye scanner);
pin pads/Key pads on doors;
smartphone access(cloud-based) / mobile access control;

Examiners report

Almost all were able to identify one alternative method and to give a good description.

(c.ii) Describe how the method identified in (c)(i) functions. [3]

Markscheme

*Award **[3 max]***

The answer to partc (ii) should match the candidate's answer to part (i). For example:

a camera is used to scan the iris / finger print is scanned;

a database stores images (scanned iris/finger print of each employee) and the rooms they are allowed access to;

computer compares the scanned image to images stored in the database;

if found, the employee is allowed to enter;

Examiners report

Almost all were able to identify one alternative method and to give a good description.

(d) Compare polling **and** interrupts as mechanisms for the swipe-card readers to interact with the central computer. [4]

Markscheme

*Award **[4 max]***

Polling requires the computer to actively interact with each swipe device;

the computer periodically checks each swipe device whether it has requested service, if it does not require servicing, the computer examines the next one, etc. If one of them requires servicing, the computer switches to running the serving program;

polling will waste processing time whilst the device is idle;

whilst:

interrupt requires the device to flag the computer;

processor receives interruption signal and services the interrupt by calling the appropriate system subroutine for interrupt processing/interrupt

handler;
and hence the computer's time is not wasted whilst the swipe device is idle;

Examiners report

It is evident that students generally have problems with compare questions, describing the issues individually without performing any comparison, as was the case here.

11. [Maximum mark: 15] 19N.1.HL.TZ0.11

A washing machine manufacturer has created its website to be viewed on standard desktop computers as well as mobile devices. The mobile browsing experience differs from desktop browsing.

(a.i) Define the term *screen resolution*. [1]

Markscheme

Award ***[1 max]***

Screen resolution is the number of pixels a screen can display (horizontally and vertically). (For example, the screen can show 1024 pixels horizontally and 768 vertically);

Examiners report

Mostly a well answered question.

(a.ii) Describe **two** issues resulting from the website being viewed on various devices, such as desktops and smartphones. [4]

Markscheme

Award ***[4 max]***

Screens that are different sizes (for example, 3–6 diagonal inches for a phone, and 9–12 diagonal inches for a tablet, size of 13–17 inches for a notebook screen, and a desktop screen size of 20–30 inches);
can still have the same screen resolution (for example, a laptop could have a 13-inch screen with a resolution of 1280 x 800 and a desktop computer could have a 17 inch monitor with the same 1280 x 800 resolution);
but physically smaller screen will not show less of the website;
the screen with the higher resolution will be able to show you more of the website because that screen has more pixels and the image will be sharper;
but elements on the screen (icons and text) will look smaller;

Most mobile displays currently have screens with fewer pixels than desktop displays and are physically smaller;
typing on small on-screen keyboards is difficult;
less precision (clicking a 12-pixel-high text link with a mouse is no problem but tapping the same link with fingers could be difficult);
there is no mouse pointer so there is no concept of "hovering" over a page element/ remaining in an uncertain state;
most modern touch screens allow zooming;
allow the user to perform gestures using one or more fingers, such as swiping/pinching;

Note*: Award marks not only for issues for a viewer but also for issues for a creator of the interface*.

Examiners report

Mostly a well answered question.

Different devices such as desktop computers and mobile devices have different operating systems.

(b) Explain the role of the operating system (OS) in terms of managing the hardware resources.

[4]

Markscheme

Award ***[4 max]***
OS controls all the activities of computer system and acts as an interface between user and hardware;

Thus the role of OS is
to keep track of who is using which resource;
to grant resource requests;
to mediate conflicting requests from different users/programs;
to allocate time to different programs or different users/ each one gets their turn to use that resource (for example printer);
to allocate space for different users, each one gets a part of the resource (for example sharing main memory/ hard disk);

Examiners report

Candidates mostly gave good responses for this question.

(c) A washing machine uses a control system.

The microprocessor controls the washing machine and its actions. To complete the wash and rinse process the user selects the program, loads the washing machine and pushes the start button.

Describe the interaction between the sensors, microprocessors and output transducers in this situation. [6]

Markscheme

Award ***[6 max]***
Award ***[2 max]*** *for evidence that*:

Sensors (input devices) detect/measure the water level;
sensors detect/measure the temperature (of the water);
sensors detect/measure the dampness/moisture level of the clothes;
sensors also detect movement of the machine's drum and other associated actions;

Sensors continuously take readings/measurements (in the context of above) and send these readings to the processor;

Award ***[2 max]*** *for evidence that*:
processor controls sensors, valves and actuators responsible for controlling the parts that clean clothes;
processor determines what actions the machine should take next;
the washing machine has been programmed/ it goes through a process of running its internal programs;

processor compares readings with pre-set values (in the context of the various sensors);
if the readings fall outside of the specified range, the processor sends a message to the output transducer to switch on/off ... (in the context of part of the washing machine);

Award ***[2 max]*** *for evidence that*:
output transducers are used for turning on and off devices that control the rest of the machine;
such as the motors that spin the tub;
or the water pump;

Example:
once the start button is pushed the washing machine begins to dump water into the drum, a sensor will detect that the water level has been reached;
based on the setting, the processor will allow the water to flow only to a predetermined level;
it will send the signal to output transducers to shut off the water;
and begin the agitation process;
once the timer tells that it is time, processor sends signal to output transducer to stop agitating;

it then begins the spin process/ removing the dirty water from the machine at the end of the spin cycle, the washing machine's processor sends signal to turn on the water pump and sucking it out of the machine;

Examiners report

A wide variety of answers and quality of answers was seen for this question covering the full mark range. Candidates who were able to fully describe the interaction between the sensors, microprocessors and output transducers and the steps involved generally scored quite highly.